Instrumental Variables

Session 1: Foundations and Mechanics

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Empirical Methods for PhD Students

Session 1 Roadmap

Outline

Three Conditions for a Valid Instrument

A valid instrument Z_i must satisfy three conditions:

1. **Relevance (First Stage):** Z_i must be correlated with the endogenous treatment D_i
   - Testable: first-stage F-statistic (rule of thumb: $F > 10$)
2. **Independence (As-Good-As-Random Assignment):** Z_i must be uncorrelated with unobserved confounders ε_i
   - Not directly testable; requires economic reasoning and robustness checks
3. **Exclusion Restriction:** Z_i affects outcome Y_i *only* through its effect on D_i
   - No direct effect; testable indirectly via placebo/falsification

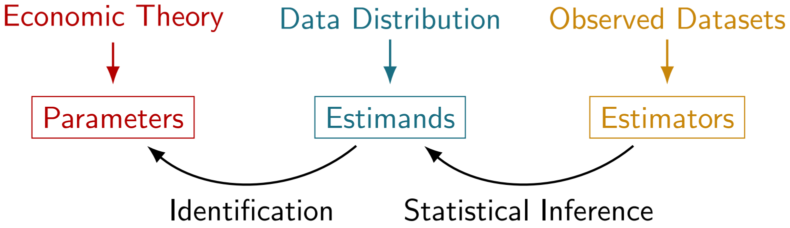
In Session 2 we add a 4th condition: **Monotonicity** (for the LATE interpretation).

Parameters, Estimands, and Estimators

Three distinct objects, not always clearly distinguished:

- Parameters come from economic (or other) models of the world
  - E.g. a potential-outcome model relating schooling to earnings
  - They set the *target* for empirical analyses: what we want to know
- Estimands are functions of the population data distribution
  - E.g. a difference in means or ratio of population regression coefficients
  - Identification: assumptions that link parameters to estimands
- Estimators are functions of observed (sample) data
  - E.g. a sample ratio of OLS coefficients
  - Estimators are random variables; statistical inference studies their distribution

The Lay of the Land



Separating identification (linking parameters to estimands) from inference (linking estimators to estimands) keeps the analysis disciplined.

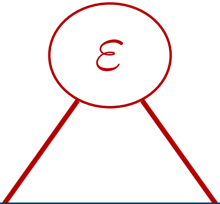
Simple causal model: $Y_i = \beta\, D_i + \varepsilon_i$, where D_i is the treatment, ε_i is the unobserved term.

In large samples, OLS converges to:

$$\beta^{OLS} = \beta + \frac{Cov(\varepsilon_i, D_i)}{Var(D_i)}$$

- OLS identifies β *only if* $Cov(\varepsilon_i, D_i) = 0$
- **Selection bias**: units with higher/lower ε_i select into treatment, so $Cov(\varepsilon_i, D_i) \neq 0$

Causal diagram:



Suppose $Z_i \in \{0, 1\}$ is a binary instrument (e.g. lottery assignment).

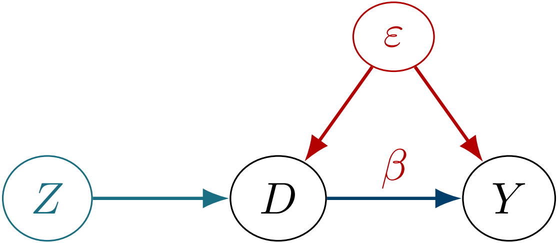
- **Randomness**: $Cov(Z_i, \varepsilon_i) = 0$ (no direct link to unobservables)
- **Exclusion**: Z_i affects Y_i only through D_i

Plugging in $\varepsilon_i = Y_i - \beta D_i$:

$$Cov(Z_i, Y_i - \beta D_i) = 0 \implies \beta = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, D_i)} \equiv \beta^{IV},$$

so long as $Cov(Z_i, D_i) \neq 0$ (relevance).

Causal diagram (IV adds Z):



No arrow from Z to ε (independence) and no arrow from Z to Y

Reduced Form and First Stage

A simple decomposition links IV back to familiar OLS regressions:

$$\beta^{IV} = \frac{Cov(Z_i, Y_i)}{Cov(Z_i, D_i)} = \frac{Cov(Z_i, Y_i)/Var(Z_i)}{Cov(Z_i, D_i)/Var(Z_i)} \equiv \frac{\rho}{\pi}$$

where ρ and π come from two OLS regressions:

$$Y_i = \kappa + \rho\, Z_i + \nu_i \quad (\text{“reduced form”})$$
$$D_i = \mu + \pi\, Z_i + \eta_i \quad (\text{“first stage”})$$

- The **first stage** shows how strongly Z_i moves D_i
- The **reduced form** shows the net effect of Z_i on Y_i
- IV scales the reduced form by the first stage to recover β

Adding Controls

If Z_i is as-good-as-random only *conditional* on covariates W_i, add them to both equations:

$$Y_i = \kappa + \rho Z_i + W_i'\phi + \nu_i$$
$$D_i = \mu + \pi Z_i + W_i'\psi + \eta_i$$

- The Frisch-Waugh-Lovell theorem tells us the effective instrument is now $\tilde{Z}_i$: the residuals from regressing Z_i on W_i
- If W_i is a set of randomization-strata dummies, $\tilde{Z}_i$ captures within-strata variation in Z_i
- Key: controls should be chosen on substantive grounds, not to maximize the first-stage F-statistic

Classic Example: Angrist (1990) Vietnam Draft Lottery

Angrist (1990, AER) estimated the earnings effect of military service using Vietnam-era draft-eligibility as an instrument.

- $Z_i \in \{0, 1\}$: randomly assigned draft eligibility (lottery number)
- $D_i \in \{0, 1\}$: actual military service
- Y_i: later-life earnings

With binary Z_i, the IV estimand simplifies to the **Wald estimand**:

$$\beta^{IV} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

- Numerator: earnings effect of draft eligibility (reduced form)
- Denominator: effect of eligibility on actual service probability (first stage)
- Finding: military service reduces early-career earnings by $\approx 15\%$

Outline

Multiple Instruments: Overidentification

When $\dim(Z_i) = L > J = \dim(X_i)$, the system is *overidentified*:

$$Cov(\tilde{Z}_i, Y_i - X_i'\beta) = 0 \implies \underbrace{Cov(\tilde{Z}_i, Y_i)}_{L\times 1} = \underbrace{Cov(\tilde{Z}_i, X_i')}_{L\times J}\underbrace{\beta}_{J\times 1}$$

- We have more moment conditions than parameters
- Any full-rank $J \times L$ matrix M gives a valid IV estimator: $\beta^{IV}(M) = \left(M\,Cov(\tilde{Z}_i, X_i')\right)^{-1} M\,Cov(\tilde{Z}_i, Y_i)$
- Choice of M trades off efficiency vs. interpretability

Two-Stage Least Squares (2SLS)

2SLS chooses $M' = \pi$ (the matrix of first-stage coefficients):

- Effective instrument: $\tilde{Z}_i^* = \tilde{Z}_i'\pi$ (residualized first-stage fitted values)
- Combine instruments according to their predictive power for $X_i \Rightarrow$ minimizes the 2SLS standard error

Two-stage interpretation:

1. *First stage*: regress X_i on Z_i and W_i; save fitted values $\hat{X}_i$
2. *Second stage*: regress Y_i on $\hat{X}_i$ and W_i

Practical warning

Never run 2SLS by hand—the second-stage standard errors are wrong. Use `ivreg2` (Stata) or `AER::ivreg()` / `fixest::feols()` (R).

2SLS as a Weighted Average of Just-Identified IVs

When there is a single treatment ($J = 1$), 2SLS has an elegant weighted-average representation:

$$\beta^{2SLS} = \sum_{\ell} \omega_\ell \, \beta_\ell^{IV}, \quad \omega_\ell = \frac{\pi_\ell^2 Var(\tilde{Z}_{i\ell})}{\sum_{\ell'} \pi_{\ell'}^2 Var(\tilde{Z}_{i\ell'})}$$

- $\beta_\ell^{IV} = \rho_\ell / \pi_\ell$ is the just-identified IV estimate using instrument ℓ alone
- Weights ω_ℓ are proportional to each instrument's contribution to first-stage variation
- Instruments with stronger first stages receive larger weights
- This "visual IV" representation (slope of ρ_ℓ on π_ℓ) is also useful for diagnosis

Overidentification Tests

Under the *constant-effects* model, all just-identified IVs identify the same β:

- If β_ℓ^{IV} differ significantly across instruments, something is wrong
- The Sargan–Hansen statistic tests $H_0 : \beta_\ell^{IV} = \beta$ for all ℓ; asymptotically $\chi^2(L - J)$ under the null

Cautions:

- Low power when individual $\hat{\beta}_\ell^{IV}$ are noisy
- Rejection does *not* tell you which instrument is invalid
- Under *heterogeneous* effects (Session 2), two valid IVs can legitimately have different estimands; the test is not interpretable
- Use the test as one input among many, not as a pass/fail criterion

Outline

If π is small relative to its standard error, the instrument is "weak":

- Rule of thumb: first-stage $F < 10$ signals potential weakness (Staiger and Stock, 1997)
- **Bias**: 2SLS is biased toward OLS in small samples; the bias is proportional to $1/F$ approximately
- **Size distortion**: standard confidence intervals may under-cover in finite samples

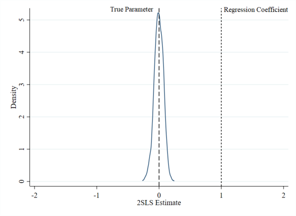
Modern guidance (Angrist and Kolesár 2022):

- In the just-identified case, 2SLS is *median-unbiased*; the SE increase tends to "cover up" the bias
- Use `weakivtest` (Stata; Montiel Olea and Pflueger 2015) for more nuanced critical values
- If F is small: consider Anderson–Rubin or JIVE as robust alternatives

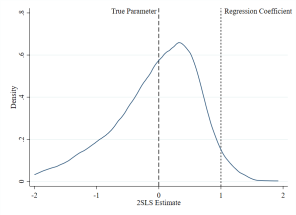
Weak Instruments: Monte Carlo

$Y_i = 0 \cdot D_i + \varepsilon_i$, $D_i = \pi Z_i + \eta_i$, $N = 100$ replications

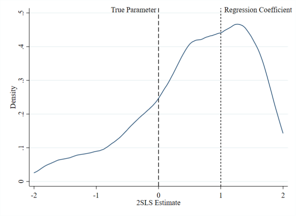
Strong ($\pi = 1$, $F \approx 100$)



Medium ($\pi = 0.1$, $F \approx 10$)



Weak ($\pi = 0.01$, $F \approx 1$)



Distribution of $\hat{\beta}^{IV}$ widens dramatically as the instrument weakens. With a very weak instrument, $\hat{\beta}^{IV}$ converges to $\hat{\beta}^{OLS}$.

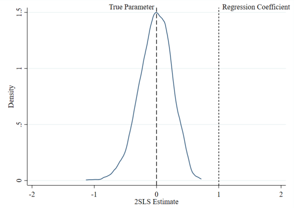
Many-IV Bias

Overidentified 2SLS faces an additional problem when there are many weak instruments:

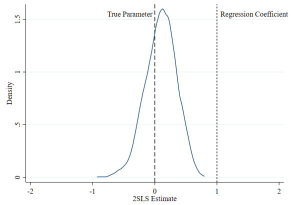
- **Symptom**: low first-stage F even though SEs are small (more IVs $\Rightarrow$ richer first stage $\Rightarrow$ better fit of D_i)
- **Mechanism**: overfitting the first stage essentially recreates the endogenous variation in D_i, biasing 2SLS toward OLS
- **Historical case**: Angrist–Krueger (1991) QoB IV interacted with state $\times$ year fixed effects created hundreds of instruments

Visualized — 100 replications:

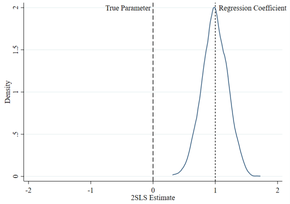
Few IVs



10 IVs



100 IVs



Practical Guidance on Instrument Strength

Prevention:

- Prefer parsimonious instrument sets; avoid mechanical many-IV settings
- If using a Bartik/shift-share IV with many industries, the *effective* number of instruments is the number of industries (addressed in Session 2)

Diagnosis:

- Report first-stage F-statistic (or Kleibergen–Paap for clustered SEs)
- Use `weakivtest` in Stata for heteroskedasticity-robust critical values
- See also Lee et al. (2020) for t-ratio-based inference

If the instrument is weak:

- Try a more informative functional form for Z_i
- Collapse to fewer instruments
- Use Anderson–Rubin confidence sets (valid regardless of strength)

Outline

Overview: Two Landmark IV Papers (Post-2010, Top-5 Journals)

We study two highly cited papers that use creative IV designs to establish causal relationships:

1. **Nunn and Qian (2014, AER)**
   "US Food Aid and Civil Conflict"
   Uses US wheat production $\times$ historical aid patterns as IV. Context: Africa and developing world.

2. **Acemoglu and Restrepo (2020, JPE)**
   "Robots and Employment: Evidence from US Commuting Zones"
   Uses Bartik shift-share IV with robot adoption in other countries. Context: US labor markets.

Both papers were cited 1,000+ times within a decade of publication and have had substantial impact on empirical methodology.

Citation

Nunn, N. and Qian, N. (2014). "US Food Aid and Civil Conflict." *American Economic Review*, 104(6): 1630–1666. (≈1,700 Google Scholar citations)

Research question: Does US food aid *cause* civil conflict in recipient countries?

Context & motivation:

- The US is the world's largest food-aid donor; annual shipments exceed $1 billion
- Theoretical ambiguity: aid may reduce conflict by alleviating poverty, or *increase* conflict by providing resources to loot or by strengthening rebel groups
- OLS correlation is unreliable: countries in conflict receive more aid (*reverse causality*); unobserved factors (instability, colonial history) drive both variables (*omitted variable bias*)

The IV: Interaction of US wheat production with a country's historical propensity to receive US food aid:

$$Z_{it} = \underbrace{\text{USWheat}_t}_{\text{supply shock}} \times \underbrace{\bar{P}_i}_{\text{historical aid share}}$$

- USWheat_t: deviations in US wheat production (driven by domestic weather and farm policy, *exogenous* to recipient conditions)
- $\bar{P}_i$: fraction of years country i received US food aid in 1971–1974 (pre-determined baseline capturing structural factors: colonial ties, Cold War alignment, proximity)

Identifying variation:

- In bad US harvest years, food-aid flows fall globally
- Countries with higher historical aid propensity experience larger fluctuations—variation that is uncorrelated with their own domestic conditions

Data:

- 125 food-aid recipient countries, annual panel 1971–2006
- **Outcome**: intra-state conflict onset & incidence (from PRIO Armed Conflict Dataset)
- **Treatment**: US food aid shipments (USAID Greenbook)
- **Instrument**: USDA wheat production $\times$ historical aid share
- **Controls**: country and year fixed effects; oil prices; US GDP growth; country-specific trends

Specification:

$$\text{Conflict}_{it} = \alpha_i + \mu_t + \beta\,\text{FoodAid}_{it} + \mathbf{W}'_{it}\gamma + \varepsilon_{it}$$

Instrumenting FoodAid_{it} with Z_{it} via 2SLS.

Level of identifying variation: US wheat production varies year-by-year; the interaction creates country-year level variation.

Key findings:

- A 10% increase in food-aid deliveries raises the probability of civil conflict onset by ≈ 1.14 percentage points (large relative to an $\approx 5\%$ base rate)
- Effects are concentrated in countries with weak initial institutions and high ethnic fractionalization
- Stronger effects for food aid that arrives *during* conflict (suggests rebel looting mechanism)

Robustness and sensitivity:

- *Placebo crops*: using soybean or corn production instead of wheat yields near-zero first stages and insignificant reduced forms—these crops are not price-supported and less likely to drive food-aid volumes
- *Alternative instruments*: using only lagged US wheat production (dropping $\bar{P}_i$ variation) gives similar results
- *Excluding major recipients*: dropping top food-aid recipients (Egypt,

Nunn & Qian (2014): Contribution and IV Lessons

Academic contributions:

- First large-N causal evidence that food aid can increase, not decrease, conflict—a "boomerang effect"
- Shapes the debate on aid effectiveness (Easterly vs. Sachs) and humanitarian intervention design
- Demonstrates that agricultural commodity shocks in donor countries can serve as valid instruments for aid flows

IV methodology lessons:

- The **interaction instrument** $Z_{it} = \text{Shock}_t \times \text{Exposure}_i$ is a workhorse design in development economics
- Validity rests on the *supply* side (US weather/farm policy unrelated to conflict in recipient countries)
- The inclusion of country FE absorbs time-invariant confounders correlated with $\bar{P}_i$; year FE absorbs global shocks
- Always test first stage separately by sub-groups and with alternative instrument definitions

Paper 2: Acemoglu and Restrepo (2020, JPE)

Citation

Acemoglu, D. and Restrepo, P. (2020). "Robots and Employment: Evidence from US Commuting Zones, 1990–2007." *Journal of Political Economy*, 128(6): 2188–2244. (≈1,800 Google Scholar citations)

Research question: What is the causal effect of industrial robot adoption on local labor markets?

Context & motivation:

- Industrial robot density rose sharply in Europe and the US from the 1990s—particularly in automotive, electronics, and metals
- Economic theory predicts robots displace routine tasks but may create complementary jobs elsewhere
- OLS is biased: industries undergoing productivity shocks may adopt robots *and* have other trends in wages/employment

The instrument: Robot adoption in other high-income countries "translated" to US commuting zones via pre-period industry shares:

$$Z_i = \sum_k \underbrace{l_{ik,1990}}_{\text{share}} \times \underbrace{\Delta\text{Robots}_{k,\text{OtherCountries}}}_{\text{shock}}$$

- **Shocks**: growth in robots per worker in industry k across Europe (France, Germany, Italy, Finland, Sweden, etc.)
- **Shares**: CZ i's employment share in industry k as of 1990 (pre-determined)

Validity argument (parallel to ADH 2013):

- Robot adoption in European industries reflects global technology improvements, not US local demand conditions
- Pre-period shares capture industrial composition that predates the robot wave—plausibly uncorrelated with contemporaneous US labor demand shocks

Data:

- 722 US commuting zones (CZs), two periods: 1990–2007
- **Treatment**: robots per 1,000 workers in each CZ (constructed from IFR data mapped to CZ industries)
- **Outcomes**: employment-to-population ratio; log wages; share in manufacturing; hours worked
- **Instrument**: exposure to foreign robot adoption via Bartik construction above
- **Controls**: demographics, routine task intensity, CZ fixed effects, census division fixed effects

Specification:

$$\Delta Y_i = \alpha + \beta\, \Delta \mathrm{Robots}_i + X_i'\gamma + \varepsilon_i$$

Instrumenting $\Delta \mathrm{Robots}_i$ with Z_i via 2SLS across two long differences (1990–2000, 2000–2007).

Key findings:

- One additional robot per 1,000 workers $\Rightarrow$ employment ratio falls by 0.18–0.34 pp and wages by 0.25–0.5%
- Roughly 400,000–670,000 jobs displaced by robots over 1990–2007
- Effects concentrated in manufacturing; no significant offsetting job creation in other sectors
- Larger effects for lower-skill, routine-task workers

Robustness and sensitivity:

- *Pre-trend test*: robot exposure does not predict 1970–1990 employment trends (no pre-existing differences)
- *Alternative shares*: using 1970 industry shares (farther from the robot wave) gives similar results
- *Excluding auto-heavy CZs*: dropping the Midwest automotive belt does not materially change estimates
- *China competition control*: conditioning on ADH-style Chinese import competition exposure leaves robot effects intact

Acemoglu & Restrepo (2020): Contribution and IV Lessons

Academic contributions:

- Provides first rigorous causal evidence that robots reduce employment and wages—fueling the automation-inequality debate
- Methodologically pairs theory (task-based model) with credible causal identification
- Spawned a large literature on technology, automation, and labor market polarization

IV methodology lessons:

- The **Bartik shift-share** design (industry shares $\times$ global shocks) is now a standard tool in regional economics—see also Borusyak, Hull, Jaravel (2022) for formal conditions
- Foreign-country shocks are a natural exogeneity source when domestic conditions are confounded
- Pre-trend and placebo tests are essential for validating the parallel-trends implicit in share-based instruments
- SSIV formal theory and China HSR illustration are covered in

Comparing the Two Papers

	Nunn & Qian (2014)	Acemoglu & Restrepo (2020)
Journal	AER	JPE
Setting	Developing countries (Africa)	US commuting zones
Treatment	US food aid flows	Robot adoption rate
IV type	Interaction (shock $\times$ exposure)	Bartik shift-share
Exog. source	US wheat production	Foreign robot adoption
Unit	Country $\times$ year	CZ $\times$ period
Key finding	Aid $\uparrow$ conflict	Robots $\downarrow$ employment & wages
Citations	$\approx$1,700	$\approx$1,800

Common thread: Both exploit variation from a *different* level (global supply shocks; foreign industry adoption) that is plausibly exogenous to the unit of interest (recipient country; US commuting zone).

Outline

Session 1 Summary

IV Foundations:

- IV validity requires relevance, independence, and exclusion
- $\beta^{IV} = \rho/\pi$ (reduced form $\div$ first stage)
- With controls: effective instrument $\tilde{Z}_i =$ residual of Z_i on W_i (Frisch–Waugh)

2SLS and Overidentification:

- 2SLS $=$ efficiency-optimal combination of multiple instruments
- 2SLS $=$ weighted average of just-identified IVs
- Sargan–Hansen test: diagnostic, not a "certification of validity"

Instrument Strength:

- $F > 10$ rule of thumb; `weakivtest` for robust critical values
- Many-IV bias: overfitting first stage recreates endogeneity

Applications (Top-5, Post-2010):

- Nunn & Qian (2014, AER): food aid $\Rightarrow$ conflict using interaction IV
- Acemoglu & Restrepo (2020, JPE): robots $\Rightarrow$ employment loss

Preview: Session 2

1. **IV Interpretation (LATE)**
   - Imbens–Angrist (1994) assumptions: monotonicity, compliers
   - Characterizing compliers; diff-in-diff IV
2. **Formula Instruments**
   - Shift-share IV: formal conditions (Borusyak et al. 2022)
   - Recentered IV: beyond shift-share (Borusyak and Hull 2023)
   - Illustration: China High-Speed Rail market access
3. **IV with Chinese Economic Context**
   - Meng, Qian, Yared (2015, *ReStud*): Great Famine IV
   - Khandelwal, Schott, Wei (2013, *QJE*): WTO quota removal

Instrumental Variables — Session 1

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Questions welcome via email.

Session 2 slides cover IV interpretation and Chinese applications.